

Dry Bean Classification Using Multi-Layer Perceptron (MLP) with K-Fold Cross Validation

Dataset

Dry Bean Dataset

The dataset consists of morphological measurements extracted from images of dry beans.

Bean Classes

- BARBUNYA
- BOMBAY
- CALI
- DERMASON
- HOROZ
- SEKER
- SIRA

Dataset Characteristics

- Multiclass Classification Dataset
- 7 Bean Varieties
- Numerical Morphological Features
- Agricultural Domain Dataset
- Supervised Learning Problem

The features describe geometric and shape-related properties of bean seeds such as:

- Area
- Perimeter
- Major Axis Length
- Minor Axis Length
- Aspect Ratio
- Compactness
- Eccentricity

- Convex Area
- Shape Factors

These attributes provide sufficient information to distinguish between different bean varieties.

Preprocessing

Label Encoding

Bean categories were converted into numerical values for neural network processing.

Example:

Class	Encoded Value
BARBUNYA	0
BOMBAY	1
CALI	2
DERMASON	3
HOROZ	4
SEKER	5
SIRA	6

Min-Max Normalization

All numerical features were normalized using Min-Max Scaling.

Formula:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Benefits:

- Removes scale imbalance
- Faster convergence
- Improved learning stability

- Better gradient updates

K-Fold Cross Validation Configuration

Parameter	Value
K-Folds	5
Validation Method	Stratified K-Fold
Shuffle	Yes
Random State	Fixed

The dataset was divided into five subsets.

For each iteration:

- 4 folds used for training
- 1 fold used for testing

The process was repeated five times.

Final performance was computed as the average across all folds.

Neural Network Architecture

The network is a custom-built Multi-Layer Perceptron implemented entirely from scratch using feedforward propagation and backpropagation.

Layer	Neurons	Activation
Input Layer	Number of Features	Input
Hidden Layer 1	64	Sigmoid
Hidden Layer 2	32	Sigmoid
Hidden Layer 3	16	Sigmoid
Output Layer	7	Sigmoid

Training Parameters

Parameter	Value
Learning Rate	0.3
Epochs	125
Optimizer	Gradient Descent
Loss Function	Sum of Squared Error
Hidden Layers	3
Cross Validation	5-Fold

Training Performance Analysis

The neural network demonstrated consistent learning across epochs.

Average Error Curve

Observation

The average error decreases steadily throughout training.

Initial Error:

- Approximately 9000+

Final Error:

- Significantly reduced after 125 epochs

Interpretation

At the beginning of training, the network has no understanding of bean characteristics.

As training progresses:

- Hidden layers learn feature relationships.
- Classification boundaries improve.
- Prediction errors reduce.

The smooth decline confirms successful optimization and convergence.

Average Training Accuracy Curve

Observation

Training accuracy increases consistently across epochs.

Interpretation

The network gradually learns:

- Shape patterns
- Area distributions
- Compactness characteristics
- Bean-specific morphological structures

The increasing accuracy demonstrates successful feature learning and improved classification capability.

Cross Validation Performance

Fold Accuracy Analysis

The K-Fold accuracy graph illustrates classification accuracy for each fold.

Interpretation

Consistent accuracy across folds indicates:

- Strong model stability
- Good generalization capability
- Low dependency on specific train-test splits

The model performs reliably across multiple subsets of the dataset.

Significance

K-Fold Cross Validation provides a more trustworthy estimate of performance than a single train-test split.

Class Distribution Analysis

Observation

The class distribution graph displays the number of samples belonging to each bean variety.

Interpretation

The distribution helps identify whether the dataset is balanced.

A reasonably balanced dataset allows the neural network to learn all classes effectively.

Significance

Balanced training data reduces prediction bias and improves fairness in classification.

Confusion Matrix Interpretation

The confusion matrix compares actual bean classes against predicted classes.

Correct Classifications

Strong diagonal values indicate successful classification of bean varieties.

The model effectively learns the morphological characteristics associated with each bean class.

Misclassifications

Errors occur when bean varieties share similar geometric characteristics such as:

- Area
- Aspect Ratio
- Compactness
- Shape Factors

These overlapping features make certain classes more difficult to distinguish.

Significance

The confusion matrix provides insight into which bean varieties exhibit the greatest similarity and classification difficulty.

Classification Report Analysis

Evaluation Metrics

The model was evaluated using:

- Precision
- Recall
- F1 Score
- Accuracy

Precision

Measures the proportion of correctly predicted bean varieties among all predictions.

Higher precision indicates fewer false positives.

Recall

Measures the proportion of actual bean samples correctly identified.

Higher recall indicates fewer false negatives.

F1 Score

The harmonic mean of Precision and Recall.

Provides balanced performance assessment.

Class-Wise Performance Interpretation

The class-wise metric graph compares:

- Precision
- Recall
- F1 Score

for all bean varieties.

Observation

Certain bean varieties achieve higher scores due to their distinctive morphological characteristics.

Other classes exhibit lower scores because they share similar geometric properties with neighboring classes.

Interpretation

The variation in class-wise performance reflects the natural overlap among bean varieties rather than model failure.

The neural network successfully captures the majority of discriminatory information contained within the feature set.

Predictions per Class Analysis

Observation

The graph shows the number of predictions assigned to each bean variety.

Interpretation

A balanced prediction distribution suggests that the model is not biased toward any specific class.

This confirms that the network learns meaningful distinctions among all bean varieties.

Storytelling and Business Insight

From Morphological Measurements to Intelligent Crop Classification

Traditionally, bean classification relies on manual inspection and expert knowledge.

However, when processing thousands of bean samples, manual classification becomes inefficient and prone to inconsistency.

This project demonstrates how machine learning can automate the process using measurable physical characteristics.

The neural network begins training without any prior understanding of bean varieties.

Through repeated exposure to:

- Shape
- Size
- Area
- Compactness
- Structural measurements

the model gradually learns to identify patterns associated with each bean type.

The decreasing error curve and increasing accuracy curve illustrate this learning process.

The successful performance across multiple validation folds confirms that the model generalizes effectively rather than memorizing the training data.

This demonstrates the practical viability of neural networks for agricultural quality-control systems.

Applications

The developed classification framework can be applied in:

Agricultural Automation

- Automated bean sorting
- Intelligent grading systems

Food Processing Industry

- Product classification before packaging
- Production line automation

Quality Control

- Consistency verification
- Defect detection support

Export Industry

- Automated crop grading
- Quality certification systems

Smart Farming

- Data-driven crop management
- Yield quality assessment

Agricultural Research

- Crop variety analysis
- Morphological pattern discovery

Overall Conclusion

This study demonstrates that a Multi-Layer Perceptron Neural Network combined with 5-Fold Cross Validation can effectively classify dry bean varieties using morphological measurements.

The model successfully learns complex relationships among geometric bean features and achieves reliable performance across multiple validation folds.

The use of:

- Feature normalization
- Backpropagation learning
- Multiple hidden layers

- Cross-validation evaluation

ensures robust and stable classification performance.

The results confirm that machine learning can serve as a practical tool for agricultural quality assessment, crop classification, and automated food-processing systems, reducing manual effort while improving consistency and scalability.